Normal Distribution

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$-\infty < x < \infty, \mu \in \mathbb{R}, \sigma > 0$$

That is, the normal distribution is parametrized by the first and second central moments.

Expected Values

$$E[X] = \mu$$

$$Var[X] = \sigma^2$$

Standardization

$$X \sim N(\mu, \sigma^2) \Rightarrow Z = \frac{X - \mu}{\sigma} \sim N(\mu_Z = 0, \sigma_Z^2 = 1)$$

Approximation for the Binomial Distribution

Requirements

$$n \geq 30, np \geq 5, nq \geq 5$$

Approximation

$$X \sim Bin(n, p) \rightarrow N(\mu = np, \sigma^2 = npq)$$

$$\frac{X - np}{\sqrt{npq}} \sim N(0, 1)$$

Warnings

This approximation fits well for cases where n is large ant npq is not too small. When $p \rightarrow 0$ (rare events), the normal distribution becomes an improper approximation since a large part of the probabilities end up to the left of $x = 0$.

In these scenarios, a better approximation is the Poisson Distribution, taking $\lambda = np$ and $P[X = k] = \frac{\lambda^k}{k!} e^{-\lambda}$.

Evaluating Normality

There are a few methods we can use to infer if a given set of observations can be modeled by the normal distribution.

Descriptive Tests

The following methods provide great insight into evaluating if a given sample fits a normal distribution. However, a more rigorous analysis is needed to properly label the distribution.

Relative Frequency Histogram

- Must be bell-shaped
- Must be symmetric in relation to the center
- Must be unimodal (single peak)

Interquartile Range (IQR)

$$IQR = Q3 - Q1$$

In a theoretical normal distribution, the IQR is approximately $1.349 * \sigma$. To evaluate a given distribution, just calculate this ratio. It should be close to 1.3.

Quantile-Quantile (Q-Q) Plot

It is the most effective descriptive tool to evaluate normality. This graph plots the actual quantiles of the sample in the y-axis and the theoretical quantiles in the x-axis.

In summary, if the sample quantiles line up perfectly with the theoretical, we have some great insight that the observed distribution must behave like a normal distribution.

Formal Statistical Tests

- Shapiro-Wilk (small to medium sample sizes)
- Kolmogorov-Smirnov with Lilliefors correction (large samples)
- Anderson-Darling (sensible to border deviations)